

Control Statements

- * ① Conditional Statements → if, else, elif
- * ② Looping Statements → for, while
- ③ Control Transfer Statements →
break, continue,
pass

Motivation :-

- ① Making decisions
- ② Repeated tasks
- ③ Handling Exceptions
- ④ Creating algorithms

Conditional Statements

if Condition:

Code block to execute

if the condition is

true

else:

Code block to execute

if the condition is

not true

More than or equal to 2 conditions

(elif)

if c1:

Code block to execute if c1 is
true

elif c2:

Code block to execute if c1
is false & c2 is true

elif c3:

code block to execute if c1 &
c2 are false & c3 is true

...

else:

code block to execute if

all the conditions are

false.

if c1: ✓ (Failed)

elif c2:

else:

if c1: ✓

if c2:

if c3:

c1

```
def multiple_of_2(num):  
    if(num%2 == 0):  
        print("Multiple of two")  
    else:  
        print("Not a multiple of 2")  
  
num = int(input("Enter a number: "))  
multiple_of_2(num)
```

c2

```
def multiple_of_2(num):  
    if(num%2 == 0):  
        return "Multiple of two"  
    else:  
        return "Not a multiple of 2"  
  
num = int(input("Enter a number: "))  
print(multiple_of_2(num))
```

original
string

reversed string

0 1 2 3 4
→ madam == madam

↳ "Palindromic String"

```

graph TD
    Loops[Loops] --> For[for]
    Loops --> While[while]
    Loops --> Arrow[ ]
    style Arrow fill:none,stroke:none
    Arrow --> End[ ]
    style End fill:none,stroke:none

```

Condition is met.

element

$$\eta = 5$$

size of list

```
for i in range(n):
```

10

20

30

40

50

$$\dot{c} = 0$$
$$\hat{c} = 1$$
$$C = 2$$
$$i=3$$
$$C = 4$$

```

for i in range(1, 6):
    for j in range(1, 11):
        print(i*j, end='\t')
    print()

```

$i = 1$ $i = 1$ _____

$J = 1$ $J = 2$ $J = 3$ $J = 4$ $J = 5$ - - - - - $J = 10$

1 2 3 4 5 10

✓
 $i = 2$

✓ $J = 1$ $J = 2$ $J = 3$ _____ $J = 10$

2 4 6 _____ 20

$i = 3$

$J = 1$ $J = 2$ $J = 3$ _____ $J = 10$

3 6 9 _____ 30

$i = 4$

$J = 1$ $J = 2$ $J = 3$ _____ $J = 10$

4 8 12 _____ 40

$i = 5$

$J = 1$ $J = 2$ $J = 3$ _____ $J = 10$

5 10 15 _____ 50

{
 1
 2
 3
 4
 5
 }

num = 1

Condition

{
 while num <= 5:
 print(num)
 num += 1

increment/decrement

decrement

outer_count = 0

1
 while outer_count < 3:
 print("Outer loop iteration", outer_count)

inner_count = 0

{
 while inner_count < 2:
 print("Inner loop iteration", inner_count)
 inner_count += 1

outer_count += 1

outer_count = ~~0~~ ~~1~~ ~~2~~ 3

Outer loop iteration 0

inner_count = ~~0~~ ~~1~~ 2

Inner loop iteration 0

Inner loop iteration 1

Outer loop iteration 1

inner_count = 0

Outer loop iteration 2

inner_count = 0

$$\underline{5!} \Rightarrow \underline{5 \times 4 \times 3 \times 2 \times 1} = \underline{\underline{120}}$$

$$\underline{0! = 1! = 1}$$